

Math 372: Homework 04

Due Friday, September 18

Problem 1. The year is 1532, and 31-year old Girolamo Cardano has been appointed mathematics chair at the University of Milan. In celebration, Cardano decides to host a dinner party. In his cellar he finds 30 bottles of wine: 8 bottles of zinfandel, 10 bottles of merlot, and 12 bottles of cabernet, all from different wineries. (Cardano only drinks red wine).

- (a) He wants to serve 3 bottles of zinfandel and the serving order is important, how many ways are there to do this?
- (b) Suppose 6 bottles are randomly selected. How many ways are there to do this?
- (c) If 6 bottles are randomly selected, how many ways are there to obtain two bottles of each variety?
- (d) What's the probability of that happening?
- (e) If 6 bottles are randomly selected, what's the probability that they are all of the same variety (i.e., all zinfandel, all merlot, or all cabernet)?

Problem 2 (Exercise 2.3.34). Computer keyboard failures can be attributed to electrical defects or mechanical defects. A repair facility currently has 25 failed keyboards, 6 of which have electrical defects and 19 of which have mechanical defects.

- (a) How many ways are there to randomly select 5 of these keyboards for a thorough inspection (without regard to order)?
- (b) In how many ways can a sample of 5 keyboards be selected so that exactly 2 have an electrical defect?
- (c) If a sample of 5 keyboards is randomly selected, what is the probability that at least 4 of these will have a mechanical defect?

Problem 3. Diego the pirate rolls two fair six-sided dice and adds the numbers together.

- (a) What is the probability that the sum is greater than 4?
- (b) Given that the sum is greater than 4, what is the probability that it is 7?
- (c) Given that the sum is even, what is the probability that Diego rolled at least one 4?
- (d) Given that at least one die shows a 3, what is the probability that the sum is 8 or 9?
- (e) Diego keeps rolling his pair of dice until the sum is either 4 or 7, then returns to his ship. What is the probability that his final roll has a sum of 4?

Problem 4. Consider randomly selecting a student at a large university. Let A be the event that the selected student has a Visa card and B be the event that the student has a MasterCard. Suppose that $\mathbb{P}[A] = .6$ and $\mathbb{P}[B] = .4$

- (a) Could it be the case that $\mathbb{P}[A \cap B] = .5$? Why or why not?
- (b) From now on, suppose that $\mathbb{P}[A \cap B] = .3$. What is the probability that the selected student has at least one of these two types of cards?
- (c) What is the probability that the selected student has neither type of card?
- (d) Describe, in terms of A and B , the event that the student has a Visa card but not a MasterCard. Then calculate the probability of this event.
- (e) Calculate the probability that the selected student has exactly one of the two types of cards.

Problem 5 (Exercise 2.3.39). A box in a supply room contains 15 compact fluorescent lightbulbs, of which 5 are rated 13-watt, 6 are rated 18-watt, and 4 are rated 23-watt. Suppose that three of these bulbs are randomly selected.

- (a) What is the probability that exactly two of the selected bulbs are rated 23-watt?
- (b) What is the probability that all three of the bulbs have the same rating?
- (c) What is the probability that one bulb of each type is selected?
- (d) If bulbs are selected one by one until a 23-watt bulb is obtained, what is the probability that it is necessary to examine at least 6 bulbs?

Problem 6 (Exercise 2.4.47). Consider randomly selecting a student at a large university. Let A be the event that the selected student has a Visa card, let B be the event that the student has a MasterCard, and let C be the event that they have an American Express card. Suppose that

$$\mathbb{P}[A] = .6, \quad \mathbb{P}[B] = .4, \quad \mathbb{P}[C] = .2 \quad (1)$$

$$\mathbb{P}[A \cap B] = .3, \quad \mathbb{P}[A \cap C] = .15, \quad \mathbb{P}[B \cap C] = .1, \quad (2)$$

$$\text{and } \mathbb{P}[A \cap B \cap C] = .08. \quad (3)$$

- (a) What is the probability that the selected student has at least one of the three types of cards?
- (b) What is the probability that the selected student has both a Visa card and a MasterCard, but not an American Express card?
- (c) Calculate and interpret $\mathbb{P}[B | A]$ and $\mathbb{P}[A | B]$.
- (d) If we learn that the selected student has an American Express card, what is the probability that she or he also has both a Visa card and a MasterCard?
- (e) Given that the selected student has an American Express card, what is the probability that she or he has at least one of the other two types of cards?

Problem 7. The prevalence of bird flu in a wild bird population is 1%. A new diagnostic test is advertised as having a false negative rate of 0.03 and a false positive rate of 0.05. To be precise, this means that

$$\mathbb{P}[\text{test is negative} | \text{bird has the flu}] = 0.03$$

$$\mathbb{P}[\text{test is positive} | \text{bird doesn't have the flu}] = 0.05$$

- (a) What is the probability that a randomly-selected bird tests positive? (*You don't need to simplify your answer.*)
- (b) Suppose that a randomly-selected bird tests positive. What is the probability that the bird has the disease? (*You don't need to simplify your answer.*)

Problem 8 (Exercise 102 on p.93). A sample of 2026 students were asked about their eye color, with the following results:

	Blue	Brown	Green	Hazel	Total
Male	370	352	198	187	1107
Female	359	290	110	160	919
Total	729	642	308	347	2026

Suppose that one of these 2026 students is randomly selected. Let F denote the event that the selected individual is female, and A, B, C and D represent the events that they have blue, brown, green, and hazel eyes, respectively.

- (a) Calculate $\mathbb{P}[F]$ and $\mathbb{P}[C]$
- (b) Calculate $\mathbb{P}[F \cap C]$. Are these events independent? Why or why not?
- (c) If the selected individual has green eyes, what's the probability that he or she is a female?
- (d) If the selected individual is female, what is the probability that she has green eyes?
- (e) What is the "conditional distribution" of eye color for females? (i.e., $\mathbb{P}[A|F], \mathbb{P}[B|F], \mathbb{P}[C|F], \mathbb{P}[D|F]$), and what is it for males? Compare the two distributions.